

CO2 AT2 – Q4. Student Registration System

Aim

To develop a Java Student Registration System using HashSet and HashMap with Generics to store unique students, assign courses, and display student data using an Iterator.

Algorithm

1. Create a Student class with student ID and name.
2. Use HashSet to store unique students.
3. Use HashMap< to assign courses to each student.
4. Read student details and add students to the HashSet.
5. Assign one or more courses to each student using the HashMap.
6. Use Iterator to traverse and display all registered students.
7. Display each student's assigned courses.
8. Display the final registration details.

Program Logic

The program uses a HashSet to prevent duplicate student records. A HashMap connects each unique student with a list of registered courses. Generics provide type safety, while an Iterator is used to traverse the HashSet and display the registration details.

Java Program

```
import java.util.*;

class Student {
    int id;
    String name;

    Student(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public boolean equals(Object o) {
        Student s = (Student)o;
        return id == s.id;
    }

    public int hashCode() {
        return Integer.hashCode(id);
    }

    public String toString() {
        return id + " - " + name;
    }
}

public class StudentRegistrationSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        HashSet<Student> students = new HashSet<>();
        HashMap<Student, ArrayList<String>> courses = new HashMap<>();

        System.out.print("Enter number of students: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            System.out.print("Enter student ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter student name: ");
            String name = sc.nextLine();

            Student s = new Student(id, name);
            students.add(s);

            System.out.print("Enter number of courses: ");
            int c = sc.nextInt();
            sc.nextLine();

            ArrayList<String> list = new ArrayList<>();

            for (int j = 0; j < c; j++) {
                System.out.print("Enter course: ");
                list.add(sc.nextLine());
            }

            courses.put(s, list);
        }

        System.out.println("\nStudent Registration Details");
        System.out.println("-----");

        Iterator<Student> it = students.iterator();

        while (it.hasNext()) {
            Student s = it.next();
            System.out.println("Student: " + s);
            System.out.println("Courses: " + courses.get(s));
        }

        sc.close();
    }
}
```

Sample Input

```
Enter number of students: 2
Enter student ID: 101
Enter student name: Anu
Enter number of courses: 2
Enter course: Java
Enter course: Database
Enter student ID: 102
Enter student name: Ravi
Enter number of courses: 2
Enter course: Python
Enter course: Networking
```

Sample Output

```
Student Registration Details
-----
Student: 101 - Anu
Courses: [Java, Database]
Student: 102 - Ravi
Courses: [Python, Networking]
```

Collections and Generics Used

Collection	Generic Type	Purpose
HashSet	Student	Stores unique student objects.
HashMap	Student, ArrayList<String>	Maps each student to their courses.
Iterator	Student	Traverses and displays students.

Important Java Concepts Used

HashSet: Stores unique students and automatically avoids duplicates based on student ID.

HashMap: Associates every student with the list of courses selected by that student.

Generics: Provides type safety, such as HashSet and HashMap>.

Iterator: Traverses the HashSet to display student registration details.

equals() and hashCode(): Make students with the same ID behave as duplicate records in the HashSet.

Result

The program successfully implements a Student Registration System using HashSet and HashMap with Generics. It stores unique students, assigns courses to each student, and displays the registration details using an Iterator.

Conclusion

This program demonstrates how Java Collections and Generics can be used to efficiently manage student records and course registrations.